

EE263 Homework 7
Summer 2026

1. Eigenvalues and singular values of a symmetric matrix. Let $\lambda_1, \dots, \lambda_n$ be the eigenvalues, and let $\sigma_1, \dots, \sigma_n$ be the singular values of a matrix $A \in \mathbb{R}^{n \times n}$, which satisfies $A = A^\top$. (The singular values are based on the full SVD: If $\text{rank}(A) < n$, then some of the singular values are zero.) You can assume the eigenvalues (and of course singular values) are sorted, *i.e.*, $\lambda_1 \geq \dots \geq \lambda_n$ and $\sigma_1 \geq \dots \geq \sigma_n$. How are the eigenvalues and singular values related?

2. More facts about singular values of matrices. For each of the following statements, prove it if it is true; otherwise give a *specific* counterexample. Here $X, Y, Z \in \mathbb{R}^{n \times n}$.

a) $\sigma_{\max}(X) \geq \max_{1 \leq i \leq n} \sqrt{\sum_{1 \leq j \leq n} |X_{ij}|^2}$.

b) $\sigma_{\min}(X) \geq \min_{1 \leq i \leq n} \sqrt{\sum_{1 \leq j \leq n} |X_{ij}|^2}$.

c) $\sigma_{\max}(XY) \leq \sigma_{\max}(X)\sigma_{\max}(Y)$.

d) $\sigma_{\min}(XY) \geq \sigma_{\min}(X)\sigma_{\min}(Y)$.

e) $\sigma_{\min}(X + Y) \geq \sigma_{\min}(X) - \sigma_{\max}(Y)$.

3. Two representations of an ellipsoid. In the lectures, we saw two different ways of representing an ellipsoid, centered at 0, with non-zero volume. The first uses a quadratic form:

$$\mathcal{E}_1 = \left\{ x \mid x^\top S x \leq 1 \right\},$$

with $S^\top = S > 0$. The second is as the image of a unit ball under a linear mapping:

$$\mathcal{E}_2 = \{ y \mid y = Ax, \|x\| \leq 1 \},$$

with $\det(A) \neq 0$.

a) Given S , explain how to find an A so that $\mathcal{E}_1 = \mathcal{E}_2$.

b) Given A , explain how to find an S so that $\mathcal{E}_1 = \mathcal{E}_2$.

c) What about uniqueness? Given S , explain how to find *all* A that yield $\mathcal{E}_1 = \mathcal{E}_2$. Given A , explain how to find *all* S that yield $\mathcal{E}_1 = \mathcal{E}_2$.

4. Principal component analysis. You are given N data points $x_1, \dots, x_N \in \mathbb{R}^n$, collected as the columns of a matrix $X = \begin{bmatrix} x_1 & \cdots & x_N \end{bmatrix} \in \mathbb{R}^{n \times N}$. Let $\bar{x} = \frac{1}{N} \sum_{i=1}^N x_i$ be the mean of the data, let $\tilde{x}_i = x_i - \bar{x}$ be the centered points, and let $\tilde{X} = \begin{bmatrix} \tilde{x}_1 & \cdots & \tilde{x}_N \end{bmatrix}$ be the centered data matrix. The (sample) *covariance matrix* of the data is

$$S = \frac{1}{N} \sum_{i=1}^N \tilde{x}_i \tilde{x}_i^\top = \frac{1}{N} \tilde{X} \tilde{X}^\top \in \mathbb{R}^{n \times n}.$$

Let $\tilde{X} = U\Sigma V^\top$ be a (full) singular value decomposition of $\tilde{X}$, with singular values $\sigma_1 \geq \sigma_2 \geq \dots \geq 0$ and left singular vectors $u_1, u_2, \dots$ (the columns of U). The u_i are called the *principal components* of the data. Throughout, “variance along a direction” means the sample variance of the data after projecting onto that direction.

a) *The direction of maximum variance.* For a unit vector $v \in \mathbb{R}^n$ ($\|v\| = 1$), the data projected onto v are the numbers $v^\top x_1, \dots, v^\top x_N$.

- i. Show that the variance of the projected data is $v^\top S v$.
- ii. Show that S is symmetric positive semidefinite, with eigenvalues $\lambda_i = \sigma_i^2/N$ and corresponding eigenvectors u_i .
- iii. Show that over all unit vectors v , the variance $v^\top S v$ is maximized by $v = u_1$, with maximum value $\lambda_1 = \sigma_1^2/N$. (So the first principal component is the direction of greatest variation.)

b) *Reduction to k dimensions, and the reconstruction error.* Approximate each point by its projection onto a k -dimensional subspace through the mean: pick $Q = \begin{bmatrix} q_1 & \dots & q_k \end{bmatrix} \in \mathbb{R}^{n \times k}$ with orthonormal columns ($Q^\top Q = I$), and set $\hat{x}_i = \bar{x} + Q Q^\top \tilde{x}_i$. Define the mean-square reconstruction error $E(Q) = \frac{1}{N} \sum_{i=1}^N \|x_i - \hat{x}_i\|^2$.

- i. Show that $x_i - \hat{x}_i = (I - Q Q^\top) \tilde{x}_i$, and use the fact that $Q Q^\top$ is an orthogonal projector to obtain

$$\underbrace{\frac{1}{N} \sum_i \|\tilde{x}_i\|^2}_V = \frac{1}{N} \sum_i \|Q Q^\top \tilde{x}_i\|^2 + E(Q), \quad \text{and} \quad V = \text{trace } S = \sum_i \lambda_i.$$

- ii. Show that $E(Q) = \text{trace } S - \text{trace}(Q^\top S Q)$, and conclude that $E(Q)$ is minimized by taking $Q = \begin{bmatrix} u_1 & \dots & u_k \end{bmatrix}$, the first k principal components, with minimum error

$$E^* = \sum_{i>k} \lambda_i = \frac{1}{N} \sum_{i>k} \sigma_i^2.$$

In words: the variance lost by keeping k components is exactly the sum of the discarded eigenvalues. You may use the result (from the lecture on symmetric matrices) that $\text{trace}(Q^\top S Q)$ over $Q^\top Q = I$ is maximized, at value $\lambda_1 + \dots + \lambda_k$, by the top k eigenvectors of S .

c) *PCA of handwritten digits.* The file `digits.json` contains $N = 1797$ grayscale images of handwritten digits, each an 8×8 image, so $n = 64$. Reading the file with `readclassjson` defines the matrix $X \in \mathbb{R}^{64 \times 1797}$ whose columns are the images (an image is displayed by reshaping a column to an 8×8 array). Carry out PCA on this data (center, then take the SVD of $\tilde{X}$).

- i. Plot the fraction of the total variance captured by the first k principal components, $\rho_k = (\sigma_1^2 + \dots + \sigma_k^2) / (\sigma_1^2 + \dots + \sigma_n^2)$, for $k = 1, \dots, n$. How many principal components are needed to capture at least 90% of the variance?

- ii. Reconstruct a few example images from their first k principal components, $\hat{x} = \bar{x} + QQ^T \tilde{x}$ with $Q = [u_1 \cdots u_k]$, for several values of k (say $k = 2, 6, 12, 24$), and display the originals next to the reconstructions. Roughly how many of the 64 components are needed before a digit is clearly recognizable? (It may also help to display the mean image and the first few principal components $u_1, u_2, \dots$ as images.)
- iii. For $k = 10$, compute the mean-square reconstruction error $\frac{1}{N} \sum_i \|x_i - \hat{x}_i\|^2$ and verify numerically that it equals $\frac{1}{N} \sum_{i>10} \sigma_i^2$, as predicted by part 2. What fraction of the total variance is lost?

5. Smoothing. We have a discrete-time signal given by $x \in \mathbb{R}^n$. We get to measure $y \in \mathbb{R}^n$, given by

$$y_i = \sum_{k=-h}^h c_k x_{i+k} + w_i \quad \text{for } i = 1, \dots, n$$

where w_i is noise. Here we use the convention that $x_i = 0$ for $i < 1$ or $i > n$. That is, y is c convolved with x plus noise. In applications, very often the effect of convolution with c is to smooth or blur x , and we would like to undo this.

The file `regl_data.json` contains c , w and x .

- a) In Julia, construct the $n \times n$ matrix such that $y = Ax + w$. Plot the singular values σ_k against k .
- b) Plot the first 6 right singular vectors of A (i.e. plot V_{ij} against i for $j = 1, \dots, 6$.) Explain what you see.
- c) Find and plot the least-squares estimate of x given y_{meas} , computing y_{meas} using c , x and w given in `regl_data.json`. Explain what happens.
- d) Many of the singular values of A are very small; this means that the measurement in the directions of the corresponding right singular vectors is being swamped by the noise.

If we believe these components are small, we can remove them from our estimate of x altogether by *truncating* the SVD of A and using the truncated SVD to compute the estimate. This is called the *truncated SVD regularization* of least-squares.

Suppose we decided only to keep the first r components. Then truncate by letting $\tilde{V}$ and $\tilde{U}$ be the first r columns of V and U , and letting $\tilde{\Sigma}$ be the top-left $r \times r$ submatrix of Σ . Then we can construct an estimator that ignores the noise components by

$$A_{\text{est}} = \tilde{V} \tilde{\Sigma}^{-1} \tilde{U}^T$$

and set

$$x_{\text{est}} = A_{\text{est}} y_{\text{meas}}$$

For values of r in 5, 10, 15, 30, 50, compute and plot the corresponding estimates of x . Explain what you see.

- e) For each r between 1 and 35, compute the norm of the error

$$\|x - x_{\text{est}}\|$$

Plot this against r . Explain what you see.

- f) Pick the ‘best’ r and plot the corresponding estimate.
- g) Another approach is to use Tychonov regularization. Find and plot the vector $x_{\text{reg}} \in \mathbb{R}^n$ that minimizes the function

$$\|Ax - y\|^2 + \mu\|x\|^2,$$

where $\mu > 0$ is the regularization parameter. Pick a value of μ that gives a good estimate, in your opinion.

- h) The regularized solution is a linear function of y , so it can be expressed as $x_{\text{reg}} = By$ where $B \in \mathbb{R}^{n \times n}$. Express the SVD of B in terms of the SVD of A . To be more specific, let

$$B = \sum_{i=1}^n \tilde{\sigma}_i \tilde{u}_i \tilde{v}_i^T$$

denote the SVD of B . Express $\tilde{\sigma}_i$, $\tilde{u}_i$, $\tilde{v}_i$, for $i = 1, \dots, n$, in terms of σ_i , u_i , v_i , $i = 1, \dots, n$ (and, possibly, μ). Recall the convention that $\tilde{\sigma}_1 \geq \dots \geq \tilde{\sigma}_n$.

- i) Find the norm of B . Give your answer in terms of the SVD of A (and μ).
- j) Find the worst-case relative inversion error, defined as

$$\max_{y \neq 0} \frac{\|AB y - y\|}{\|y\|}.$$

Give your answer in terms of the SVD of A (and μ).